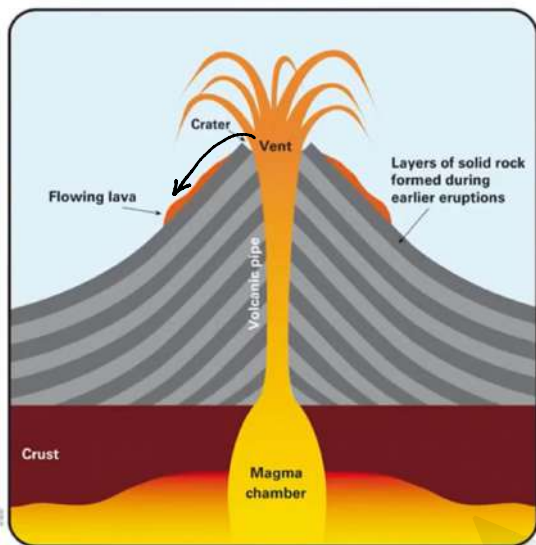


ROCKS, CONTINENT AND OCEANS



Petrology: Study of rocks



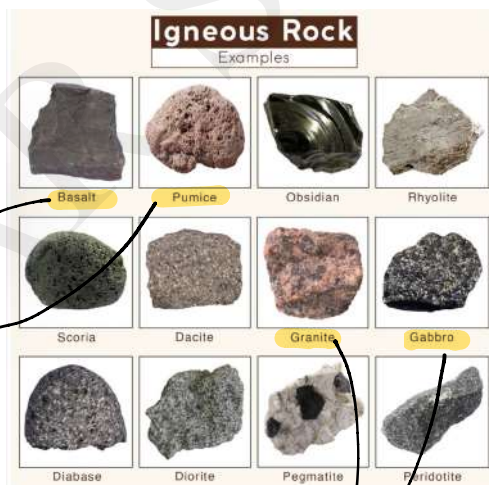
Exogenous agents act upon
eg: wind, water

How are rocks formed?

Igneous Rocks: formed when magma cools and solidifies

Types:

1. Intrusive: solidifies inside
2. Extrusive: solidifies outside



Extrusive igneous
rock

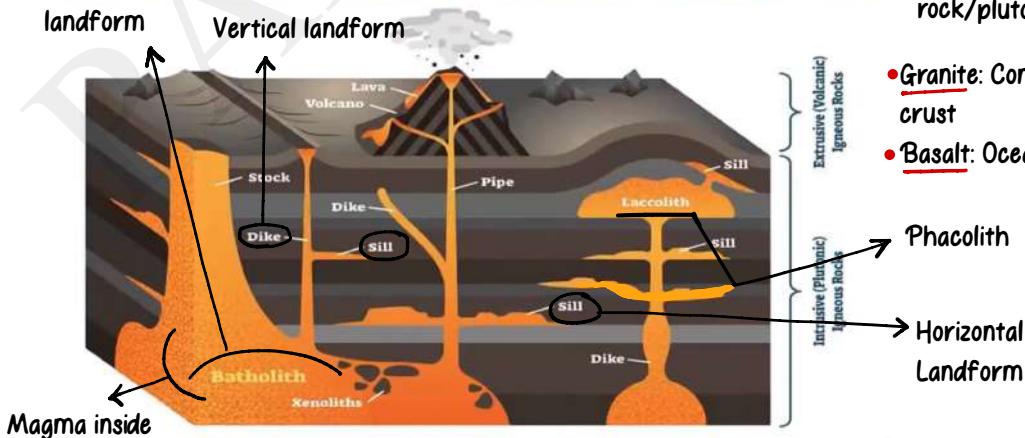
Intrusive igneous
rock/plutonic rocks

- Granite: Continental crust
- Basalt: Oceanic crust

PLUTONIC BODIES

Domed shaped
landform

Vertical landform



Sedimentary Rock: Sediments are broken, transported, and deposited

- They exist in layers/strata
- In sedimentary compaction takes place – Lithification
- Fossils are found in it

Types:

1. Formed mechanically, eg: Sandstone, limestone and shale
2. Formed organically, eg: chalk, limestone, coal
3. Formed chemically, eg: Limestone, halite



Metamorphic Rock: These rocks are formed by recrystallisation and reorganisation of materials within the original rocks

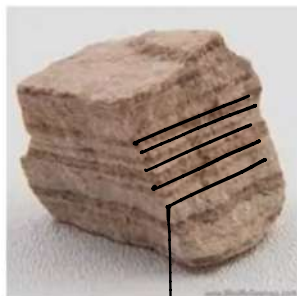
Igneous + Sedimentary → Metamorphic

↓
 Pressure
 Volume
 Temperature

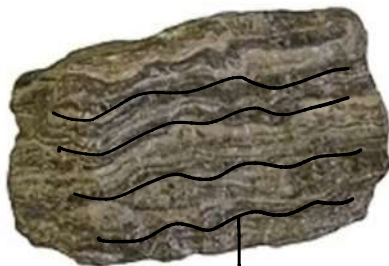
} change in forms → Metamorphosis

Types:

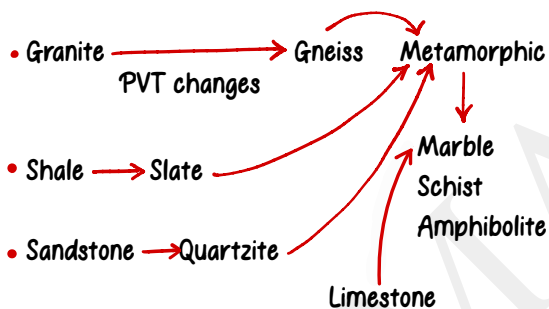
1. Thermal Metamorphism: metamorphic rocks formed due to a sudden temperature change
2. Dynamic Metamorphism: metamorphic rocks formed without any chemical change



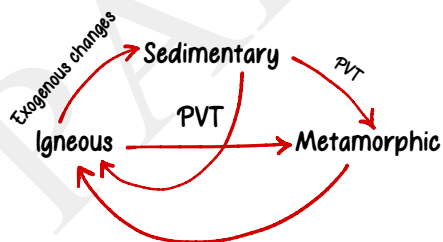
Alternate dark and light bands
called **banding**



Lines formed called **Lineation**



Rock Cycle



Volcano

Types:

1. Cinder
2. Composite: most viscous lava
3. Shield: low viscosity lava
4. Caldera: most explosive lava, collapses on itself

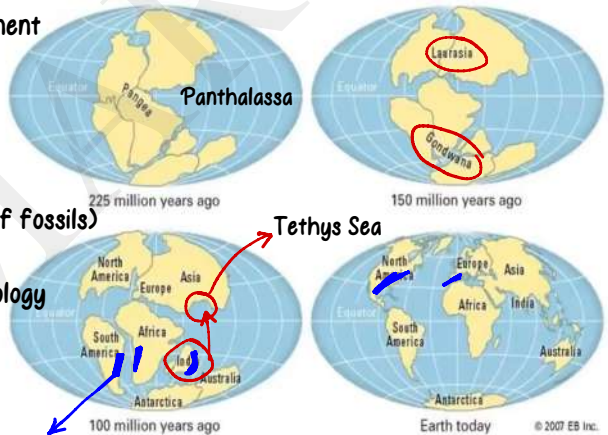
Continents and Oceans

- Alfred Wegener: Gave Continental Drift Theory, 1912

All of the modern-day continents had previously been clumped together in a supercontinent called **Pangaea** and the water body is called **Panthalassa**

→ Evidences:

- Jig Saw fit
- Fossils deposits: Palaeontology (study of fossils)
- Placer deposits



Study of rock: Petrology

Types of rocks

Soft

eg: Talc

Hard

eg: Diamond

Convection cells

Due to

Residual heat

Radioactive decay

Two main sources of heat within the Earth

- Continental drift due to (as assumed by Alfred Wegener)

1. Tidal force
2. Polar fleeing force

But it occurs due to development of **convection cells**



• Decreasing order of Continents and Oceans

Area wise

Asia
Africa
North America
South America
Antarctica
Europe
Australia

Population basis

Asia
Africa
Europe
North America
South America
Australia
Antarctica

Mariana Trench deepest point: Challenger deep

deepest ocean

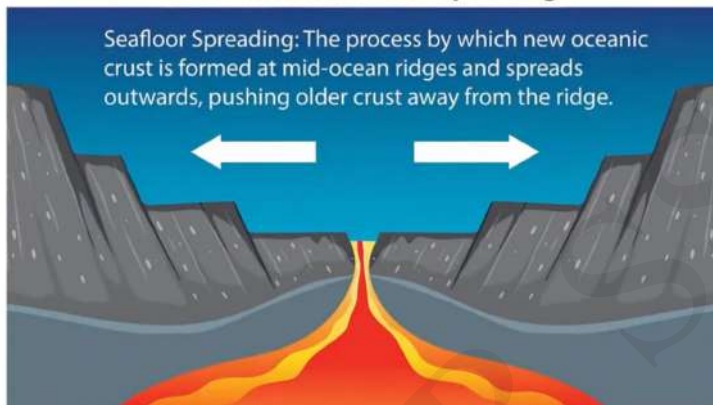
Oceans Order

P: Pacific Ocean
A: Atlantic Ocean (S-shape)
I: Indian Ocean
S: Southern (Atlantic)
A: Arctic

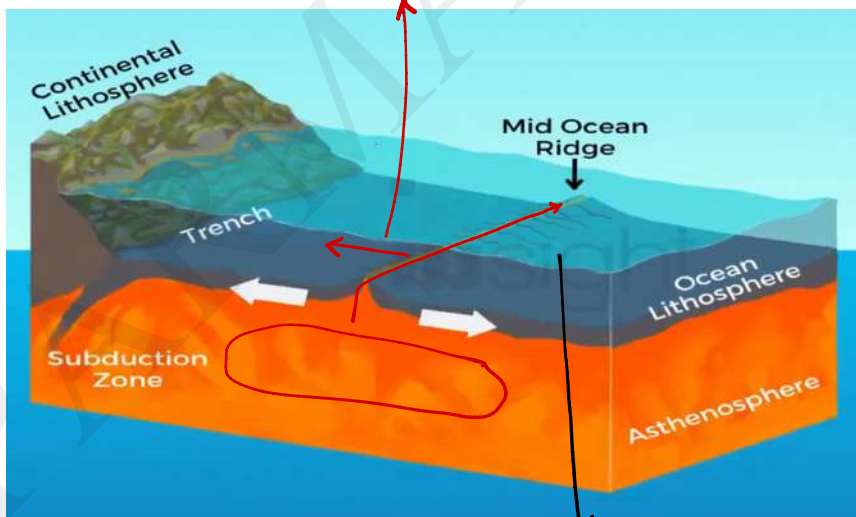
Busiest ocean

Sargasso Sea (brown algae
Sargassum is seen here) -
borderless sea

The Process of Seafloor Spreading



The age of oceanic rocks increases as you move away from the mid-ocean ridge



Oceans has more relief features than continents (more diversity)

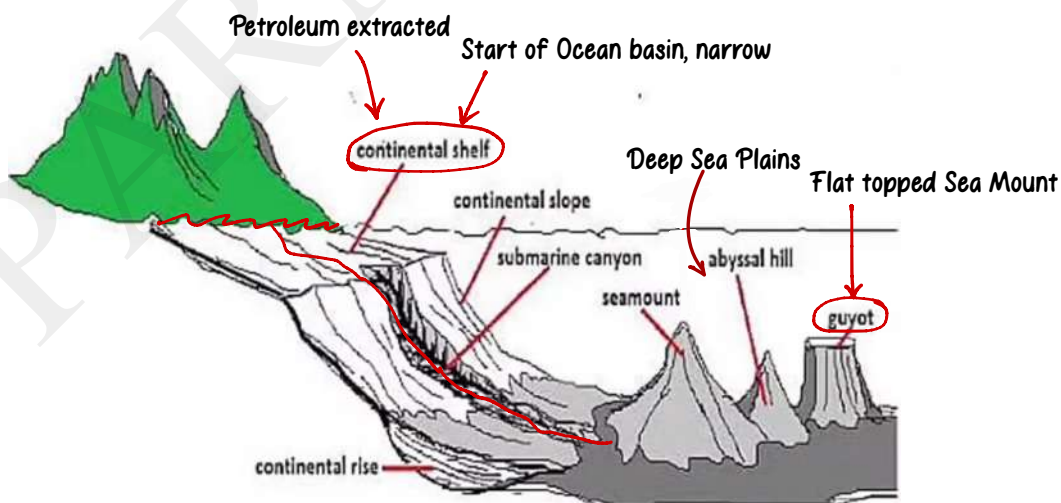
- Harry H. Hess gave seafloor spreading theory, 1962



created due to divergent
plate boundary

Mid-Atlantic Ridge

Longest and largest
landform on Earth





- **Minor relief feature: Atoll, sea mount, guyot**

Corals: they are sea organisms, known as **Rainforest of Sea**

- **Exists in symbiotic relationship with Zooxanthellae algae**

Makes food for corals

Secretes CaCO_3 that provides protection to Zooxanthellae algae

- Corals exists in colony
- Favourable conditions:
 1. Saline water (cannot survive in fresh water)
 2. Sunlight
 3. Clear water
 4. Temperature: Moderate temperature $30-35^\circ\text{C}$

- **Barrier Reef: Great Barrier Reef in Australia (largest)**

- Coral bleaching: when water is too warm, corals will expel the algae (Zooxanthellae) living in their tissues causing the corals to turn completely white

↓
due to climate change



→ Relief Feature of Oceans

Major

Shelf → Slope → Rise → Abyssal plain

Minor

• Seamount

• Guyot: flat-topped sea mount